

Experiment No: 8

Title : • Study & Implementation of different types of constraints

Objective:

To practice and implement constraints

Theory:

CONSTRAINTS:

Constraints are used to specify rules for the data in a table. If there is any violation between the constraint and the data action, the action is aborted by the constraint. It can be specified when the table is created (using CREATE TABLE statement) or after the table is created (using ALTER TABLE statement).

1. NOT NULL: When a column is defined as NOTNULL, then that column becomes a mandatory column. It implies that a value must be entered into the column if the record is to be accepted for storage in the table.

Syntax:

```
CREATE TABLE Table_Name (column_name data_type (size) NOT NULL, );
```

Example:

```
CREATE TABLE student (sno NUMBER(3)NOT NULL, name CHAR(10));
```

2. UNIQUE: The purpose of a unique key is to ensure that information in the column(s) is unique i.e. a value entered in column(s) defined in the unique constraint must not be repeated across the column(s). A table may have many unique keys.

Syntax:

```
CREATE TABLE Table_Name(column_name data_type(size) UNIQUE, ....);
```

Example:

```
CREATE TABLE student (sno NUMBER(3) UNIQUE, name CHAR(10));
```

3. CHECK: Specifies a condition that each row in the table must satisfy. To satisfy the constraint, each row in the table must make the condition either TRUE or unknown (due to a null).

Syntax:

```
CREATE TABLE Table_Name(column_name data_type(size) CHECK(logical
expression), ....);
```

Example:

```
CREATE TABLE student (sno NUMBER (3), name CHAR(10),class
CHAR(5),CHECK(class IN('CSE','CAD','VLSI')));
```

4. PRIMARY KEY: A field which is used to identify a record uniquely. A column or combination of columns can be created as primary key, which can be used as a reference from other tables. A table contains primary key is known as Master Table.

- ✓ It must uniquely identify each record in a table.
- ✓ It must contain unique values.
- ✓ It cannot be a null field.
- ✓ It cannot be multi port field.
- ✓ It should contain a minimum no. of fields necessary to be called unique.

Syntax:

```
CREATE TABLE Table_Name(column_name data_type(size) PRIMARY KEY,
....);
```

Example:

```
CREATE TABLE faculty (fcode NUMBER(3) PRIMARY KEY, fname
CHAR(10));
```

5. FOREIGN KEY: It is a table level constraint. We cannot add this at column level. To reference any primary key column from other table this constraint can be used. The table in which the foreign key is defined is called a **detail table**. The table that defines the primary key and is referenced by the foreign key is called the **master table**.

Syntax: CREATE TABLE Table_Name(column_name data_type(size)

```
FOREIGN KEY(column_name) REFERENCES table_name);
```

Example:

```
CREATE TABLE subject (scode NUMBER (3) PRIMARY KEY, subname  
CHAR(10),fcode NUMBER(3), FOREIGN KEY(fcode) REFERENCE faculty );
```

Defining integrity constraints in the alter table command:

Syntax: **ALTER TABLE** Table_Name **ADD PRIMARY KEY** (column_name);

Example: **ALTER TABLE** student **ADD PRIMARY KEY** (sno);

(Or)

Syntax: **ALTER TABLE** table_name **ADD CONSTRAINT** constraint_name
PRIMARY KEY(colname)

Example: **ALTER TABLE** student **ADD CONSTRAINT** SN **PRIMARY KEY**(SNO)

Dropping integrity constraints in the alter table command:

Syntax: **ALTER TABLE** Table_Name **DROP** constraint_name;

Example: **ALTER TABLE** student **DROP PRIMARY KEY**;

(or)

Syntax: **ALTER TABLE** student **DROP CONSTRAINT** constraint_name;

Example: **ALTER TABLE** student **DROP CONSTRAINT** SN;

6. DEFAULT : The DEFAULT constraint is used to insert a default value into a column. The default value will be added to all new records, if no other value is specified.

Syntax:

```
CREATE TABLE Table_Name(col_name1,col_name2,col_name3  
DEFAULT '<value>');
```

Example:

```
CREATE TABLE student (sno NUMBER(3) UNIQUE, name CHAR(10),address  
VARCHAR(20) DEFAULT 'Aurangabad');
```

LAB PRACTICE ASSIGNMENT:

1. Create a table called EMP with the following structure.

Name Type

EMPNO NUMBER (6)
ENAME VARCHAR2 (20)
JOB VARCHAR2 (10)
DEPTNO NUMBER (3)
SAL NUMBER (7,2)

Allow NULL for all columns except ename and job.

2. Add constraints to check, while entering the empno value (i.e) empno > 100.
3. Define the field DEPTNO as unique.
4. Create a primary key constraint for the table(EMPNO).
5. Write queries to implement and practice constraints.